

Backtest of ICES Advice

Executive summary — six Northeast Atlantic stocks

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Sea++

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What was done

We compared two counterfactual harvest policies with the latest ICES assessments for six Category 1 stocks — Irish Sea cod and whiting, Celtic Sea cod, whiting and haddock, and Northeast Atlantic mackerel — from 2005 to each stock’s terminal assessment year. The same operating-model biology and stock-recruitment residuals are used in both simulations; only the harvest policy differs. The run is **deterministic**. This draft is **one SRR branch**, not the full operating-model set.

Three trajectories: **Historical** (latest WGCSE assessment); F_{target} (**open loop**: a simple projection at constant ICES SAG F_{MSY} from 2006, lag 1; no feedback; not the yearly advice-sheet F); and **Advice rule (AR)** (**closed loop**: an MSE with feedback control; hcrICES revises F from SSB each year, $F = F_{\text{target}}$ when $\text{SSB} \geq \text{MSY } B_{\text{trigger}}$, reduced toward B_{lim} ; true OM SSB, $\text{err} = \text{NULL}$; no implementation error; one-year lag). The constant- F projection is the no-feedback benchmark that isolates fishing at F_{MSY} from following the HCR — they can disagree (Irish Sea cod historical $F \approx 0.07 \times F_{\text{target}}$, so fishing at F_{target} lowers SSB; Celtic Sea stocks often sit below $\text{MSY } B_{\text{trigger}}$, so the HCR fishes well below F_{target}). Historical figures and Table 1 use **bh3** (R_0 prior from the hockey-stick); Future projections and Table 2 use the same branch. HCR inputs come from **bh1**: unconstrained Beverton–Holt (`bevholm1SV`); steepness is estimated, not fixed. OM shading (F_{MSY} , B_{MSY} , R_0) is **bh3**. The 20-year Future window uses mean biology and mean recruitment — longer than the ICES 1–2 year advice forecast, and not a stochastic forecast.

ICES does not report B_{MSY} . Advice uses F_{MSY} , B_{lim} and $\text{MSY } B_{\text{trigger}}$ (in practice usually B_{pa}); B_{MSY} is an implicit EqSim output and is not published, so using the trigger as a surrogate over-classifies stocks as “at MSY” (Winker et al., Fish and Fisheries 2026). This pack therefore keeps ICES $\text{MSY } B_{\text{trigger}}$ as the HCR input and takes B_{MSY} from the OM equilibrium (same SRR as Historical and Future). An OM is also required because EqSim, as used in ICES benchmarks, is largely an open-loop projection at constant F and does not simulate MSE feedback; EqSim F_{MSY} is not precautionary relative to a shortcut MSE on an OM (Winker et al., ICES J. Mar. Sci. 2025, fsaf204).

Historical

All six stocks are below $\text{MSY } B_{\text{trigger}}$ in the terminal year. Realised $\bar{F}$ exceeds ICES F_{target} except in Irish Sea cod. Once SSB is above the trigger, the AR sets $F = F_{\text{target}}$ (about a one-year lag). The contrast that matters is how far each stock sits below the trigger, and how much SSB the AR gains or loses relative to history (Δ in Table 1).

Irish Sea cod is the counter-example. Historical $\bar{F}$ is already well below F_{target} ($F/F_{\text{target}} \approx 0.07$),

Table 1: Terminal-year $SSB/MSY B_{trigger}$ (Current). Historical and F_{target} / AR trajectories = bh3; HCR inputs from bh1. Under F_{target} , $F/F_{target} = 1$ by construction. Δ is AR minus Historical.

Stock	$SSB/MSY B_{trigger}$			Hist F/F_{target}	Δ
	Hist	F_{target}	AR		
Irish Sea cod	0.54	0.27	0.33	0.07	-0.21
Irish Sea whiting	0.51	4.94	4.95	5.25	+4.45
Celtic Sea cod	0.02	4.22	4.23	4.44	+4.21
Celtic Sea whiting	0.22	0.40	0.49	1.42	+0.27
Celtic Sea haddock	0.63	1.21	1.21	1.78	+0.58
NE Atlantic mackerel ^a	0.77	0.77	0.77	1.39	0.00

^a Mackerel $MSY B_{trigger}$ is the SAG field (equal to B_{pa} in the latest SAG row), not the LTMP/ B_{lim} convention on the ICES advice sheet.

so fishing at F_{target} **lowers** terminal SSB relative to history. That is the opposite of the other depleted gadoids, where the AR would have cut F . Two mechanisms can produce a realised F this far below the ICES target; this backtest does not yet separate them: mixed-fishery constraint (cod as a recovery / choke stock) versus a mismatch between ICES F_{target} and operating-model F_{MSY} . An advised-catch series would distinguish those; it is not in this draft.

Constant F_{target} rebuilds **Irish Sea whiting** and **Celtic Sea cod** to several times $MSY B_{trigger}$ (terminal $SSB/MSY B_{trigger} \approx 4.9$ and 4.2). For those stocks historical F was several times F_{target} ; putting the rule in force is a large cut in F , and SSB follows.

Future (20 years)

Both series then follow the ICES HCR for 20 years from their terminal states (bh3; mean recruitment). Reaching ICES $MSY B_{trigger}$ does not imply operating-model B_{MSY} .

Celtic Sea whiting: ICES $MSY B_{trigger}$ sits materially above OM B_{MSY} on this branch. SSB never reaches $MSY B_{trigger}$ in 20 years from either start, but it does reach OM B_{MSY} in about 16–18 years. A successful rebuild against the OM yardstick can remain “below trigger” on the ICES scale.

Celtic Sea cod: from history, $MSY B_{trigger}$ in about 9 years; OM B_{MSY} is not reached in 20 years from either start. The Advice-rule start is already above the trigger, but still short of OM B_{MSY} . Rebuild to the ICES trigger and rebuild to MSY biomass are different targets.

Irish Sea whiting reaches $MSY B_{trigger}$ in 2 years from history and B_{MSY} in 7; the AR start is already above both. **Irish Sea cod** reaches B_{MSY} first (6–7 years) because that biomass sits below ICES $MSY B_{trigger}$ (8 years). **Mackerel** is already above OM B_{MSY} while still below the (SAG) trigger.

Caveats and next

This is one SRR branch. Historical and Future trajectories are bh3; HCR parameters (ICES F_{target} , $MSY B_{trigger}$, B_{lim}) stay at SAG values on bh1. OM shading is from the simulated bh3 branch. There is no estimation error, no implementation error, and no Monte Carlo $P(SSB < B_{lim})$. Mackerel $MSY B_{trigger}$ is not on the same published basis as the gadoids.

Next: other SRR branches (bh2; Ricker rk1–rk3); an advised-catch series to separate choke/recovery TAC from an OM/ F_{target} mismatch; and stochastic replicates for $P(SSB < B_{lim})$.